



ResMed



Integrator Authentication Guide

English

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1. Overview of the authentication process

ResMed has upgraded to a new authentication and authorization mechanism for integrations that follows the standard OAuth client credentials flow.

The following steps outline the integration process with AirView™ Exchange (AVX) using the new authentication and authorization flow.

1. ResMed works with you to get all the required inputs to create or migrate your integrator account.
2. ResMed creates your integrator account in the test environment.
3. You receive credentials that you can use for testing in the test environment.
4. After you complete a successful test in the test environment and are certified and ready for production integration, ResMed sends you production credentials and URLs.
5. You authenticate your integration in production.
6. When your certificates expire after one year, ResMed will coordinate with you and send you renewed certificates.

Authenticate with AVX

To authenticate and gain access to AVX you must complete the following steps.

1. Use a `client_id` and `client_secret` with ResMed's authorization server to obtain an access token.

Note: Access tokens expire after a period of time (currently one hour, but ResMed may change this in the future). After an hour, you must reauthenticate with the authorization server to obtain a new access token to make API calls at the gateway.

2. Use the access token to create a mutual TLS connection, client certificate and private key (provided by ResMed) with ResMed's API gateway to authenticate to make API calls.

Required inputs

To migrate your integrator account to the new authentication and authorization flow, we require the following inputs:

- **Your company name:** The name of your company.
- **Your AirView integrator name:** Your integrator name as you want it to appear in AirView. This may be the same as your company name but could be different if you have multiple integrator accounts.
- **Integration email:** We recommend providing a permanent email address that is tied to more than one person.
 - Your credentials and certificate will be delivered securely to this email address (this includes renewed/new certificates when certificates expire).

- **AirView organization names (optional):** If you only want certain AirView organizations to be able to see you as an integrator, you'll want to provide the specific AirView organizations that should have access to add you as an integrator.
 - This is only required if you are a new AVX integrator.
 - This could be your company if you are a direct customer integrator.

Work with your ResMed contact to share these inputs so we can create and send you the credentials and the needed URLs to access and integrate with AVX.

To use notifications to alert you when new data or updates are available for your patients, we need the following additional fields:

Note: If you are an existing integrator that uses notifications, then no action or inputs are needed.

- **Notification URL:** The URL to which ResMed sends the notification.
- **Notification username:** A valid username to access your endpoint.
- **Notification password:** A valid password correlated with the username to access the endpoint.
- **Notification thread count:** The count of concurrent threads for receiving notifications at the same time.

2. Client credentials

After ResMed adds you into the OAuth flow as an integrator, you will receive an email at the provided address. Use the link in the email to access the files, which have the credentials, certificate and private key required to use the APIs. To download the required files, complete the following steps:

1. Click the link in the email to view the BlackBerry® Workspace login page.
2. Create a login to access the files. If the login page is blocked by your IT organization, contact your IT department to gain access or temporary access to the page. ResMed's security team recommends sharing credentials through the secured method of BlackBerry Workspaces.
3. After you create the login, you can access and download the following files:
 - **client_credentials.txt:** This contains the following:
 - **client_id:** Use this to authenticate with the authorization server.
 - **client_secret:** Use this to authenticate with the authorization server.
 - **integrator_id:** Your unique identifier as an integrator. This is not required for any API calls, but just as a reference for you and for ResMed to know.
 - **certificate_serial_number:** The serial number/identifier of your new certificate.
 - **authorization_server_url:** The URL for using the `client_id` and `client_secret` to get an access token.
 - **api_base_url:** The URL for the API gateway to connect and make API calls.
 - **NOTE:** When making API calls, the API endpoint looks like this: https://<host>/<app>/api_path.
 - `<app>` is "exchange" (without quotation marks).
 - The `api_base_url` includes the <https://<host>> part of the API endpoint. You will need to add the `<app>/` part before the rest of the API path.
 - **client.pem:** This file contains your certificate.
 - **client.key:** This file contains your private key.
 - **AVX_API_Suite.postman_collection.json:** This file is used to quickly test connectivity to the API gateway and authorization server. This will be shared via your blackberry workspace folder.

The following text is an example of the `Client.txt` file:

```
client_id: "sample_client_id"

client_secret: "sample_client_secret"

integrator_id: "sample_integrator_id"

certificate_serial_number: "sample_certificate_serial_number"

authorization_server_url: https://sample-authorization-server-url

api_base_url: https://sample-api-base-url
```

The following text is an example of the `Client.pem` file:

```
-----BEGIN CERTIFICATE-----
MIIDqTCCApGgAwIBAgIQPGZM2eCAK4S7ATMb/aUpu9TWG1RYkxqBuI8SLTlu58m6k8UUi4n
YDVQQGEwJVUzEPMA0GA1UECgwGUmVzTWVkbWQ4wDAYDVQQIDAUVUZXhhczEOMAwGA1UEBwwF
UGxhbm8wHhcNMjMwNzAzMjAwNjI2WhcNMjMwNzAyMjAwNjI2WjBlMQwwCgYDVQQDDANjb2
0xCzAJBgNVBAYTA1VTMQswCQYDVQQIDAJDQENMAwGA1UEBwwEY2l0eTEPMA0GA1UECgwG
cmVzbWVkbMRswGQYJKoZIhvcNAQkBFgxnYmNAdGVzdC5jb20wggeiMA0GCSqGSIb3DQEBAQ
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M4iqS3nCXsKJfkRwku4tQ7OvvoW/KUpTB1Xbouz47XvdgTwIptHVUfPSPVltZQi4sn5pKk
p4pCpn8xRtrL8ctb+Pwz2mtnWu8Iwty3XH3vMn+CmeW3fZkLwz9QbPjWUKfquX447PjkYc
B6BQCzTLZONK23Z4hIZ0bY12vdzlyhisg5ElUrxFGY9s0o3WC5jMTqsgwzsymDTHNGQtWD
fNTXOKSeAVXcuUQG2aqVSfjMQGqfQdt3YSf5exF6xTbitbYBOK2rLZaCZgK4Q8KLHIXTah
wFxjMR7AgMBAAGjfDB6MAkGA1UdEwQCMAAwHwYDVR0jBBgwFoAU62i14hJ84Lo48t/115E
SKWY2oGowHQYDVR0OBBYEFEPJBEEmnt0iGu08G395eXsBjXVoMA4GA1UdDwEB/wQEAwIFo
DAdBgNVHSUEfjAUBggrBgEFBQcDAQYIKwYBBQUHAWIwDQYJKoZIhvcNAQELBQADggEBACu
j9IfPcicqxm1Pdb6niDAFYX7B7JybQuVK0kvjBE0WoSn/a59iUWDreg9R9Eg3czP1VEnVm
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5PjDpqiGn4SZ9qFWMVckgufoYV9g7DcKxrpEdv1YjSLCRaDyA+FrwRTvDejbTOMehmDXgl
kXXIUt4eaoelMmTV4y84CLQHeLcKUthq1dpiMYYCfYVEwoP9Yj17133D82rnKbtdxDNiqh
OBLv9lpyD7NCWzOYVS4S0x1Hw+OEucIOjd8wBDZav20opy1jobpYivM50fDQ=-----END
CERTIFICATE-----
```

The following text is an example of `Client.key` file:

```
-----BEGIN PRIVATE KEY-----
SAMPLE PRIVATE KEY CHARACTERS
-----END PRIVATE KEY-----
```

Getting the Access Token

Before making an API call, you will need to retrieve the access token. To retrieve the access token, you will need to make an API call to the `authorization_server_url` and use the client id and client secret. A successful response will give you a valid access token.

Responses will look something similar to the following examples:

Success Response Example:

(200 OK response)

```
{
  "token_type": "Bearer",
  "expires_in": 3600,
  "access_token": "8SLTlu58m6k8UUi4nM4iqS3nCXsKJfkRwku4tQ7OvvoW/KUpTB1Xboua
IptHVUfPSPVltZQi4sn5pKkp4pCpn8xRtrL8ctb+Pwz2mtnWu8Iwty3XH3vMn+CmeW3fZkLwz9QbP
asd89SLTlu58m6k8UUi4nM4iqS3nCXsKJfkRwku4tQ7OvvoW/KUpTB1Xboua +CmeW3fZazik9d",
  "scope": "default"
}
```

Failure Response Examples:

(400 Bad request response) invalid client id:

```
{
  "errorCode": "invalid_client",
  "errorSummary": "Invalid value for 'client_id' parameter.",
  "errorLink": "invalid_client",
  "errorId": "oaeZnfg68moS8-Jl3uZtvnMCg",
  "errorCauses": []
}
```

(401 Unauthorized) invalid client secret:

```
{
  "error": "invalid_client",
  "errorDescription": "The client secret supplied for a confidential client is invalid"
}
```

Authenticating at the Gateway

To authenticate at the gateway and make an AVX API call, you'll first need your access token, your client certificate, and key. Use those things to make an API call at the various API endpoints. A successful response will give you access to make the API call. Any failure should result in a 403 forbidden error. For security reasons, little detail is shared on failure reasons.

Actual responses may not look identical to the examples below but should look similar.

Successful API Response Example:

(200 OK response)

```
{
  "Response": {
    "responseCode": "1002000",
    "responseDesc": "SUCCESS",
    "availableBillingPlans": [
      {
        "id": "87654321-11AA-22BB-33CC-1234567890AA",
        "name": "Card only",
        "availableStartDate": "2014-01-01"
      },
      {
        "id": "87654321-11AA-22BB-33CC-1234567890AA",
        "name": "Global Everything Plan",
        "availableStartDate": "2014-01-01",
        "availableEndDate": "2020-12-30"
      }
    ]
  }
}
```

Failure Response Examples:

(403 Forbidden response) Note: these could be for a variety of reasons including an expired access token.

```
{  
  "Message": "User is not authorized to access this resource with an  
explicit deny"  
}
```

```
{  
  "Message": "Forbidden"  
}
```


3. Test connectivity and authentication with Postman

To simplify testing authentication and connecting to the gateway, we created a Postman® collection with AVX API calls for you to perform an end-to-end integration test after you receive your new client credentials and the new URLs.

Note: These credentials work only with the associated URLs you are given. Test environment credentials are for the test environment and production credentials are only for production.

The Postman scripts execute a sequence of API requests against AVX's public API. If you successfully complete these requests, then your initial authentication is successful. Failures should be inspected and investigated further.

Prerequisites

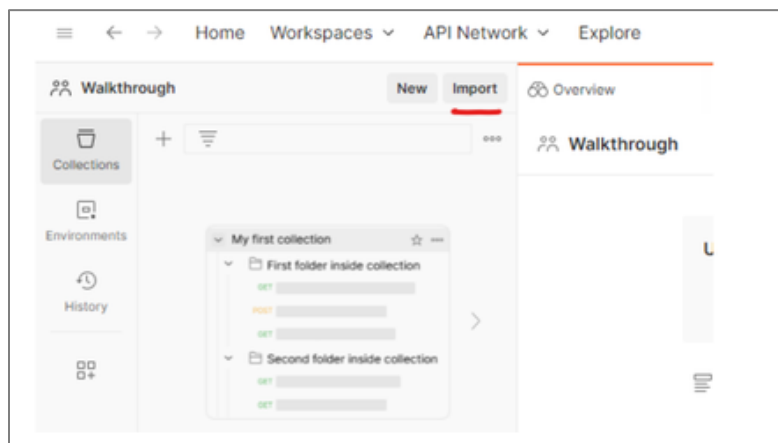
To test your integration, complete the following steps:

1. Download and install Postman [here](#).
2. Save the `AVX_API_Suite.postman_collection.json` file to an accessible local directory. This file is in your BlackBerry Workspace.
3. Access URLs and credentials through the link in the email you received from the Customer team. These should include:
 - the AVX API URL and Authorization server URL
 - your `client_id` and `client_secret`
 - the certificate and certificate key.

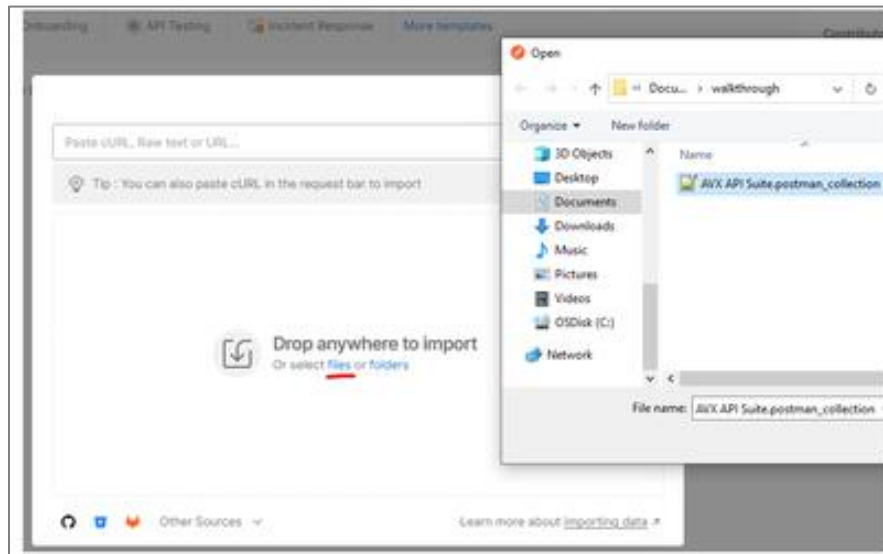
Import the collection

With Postman installed, complete the following steps to import the collection:

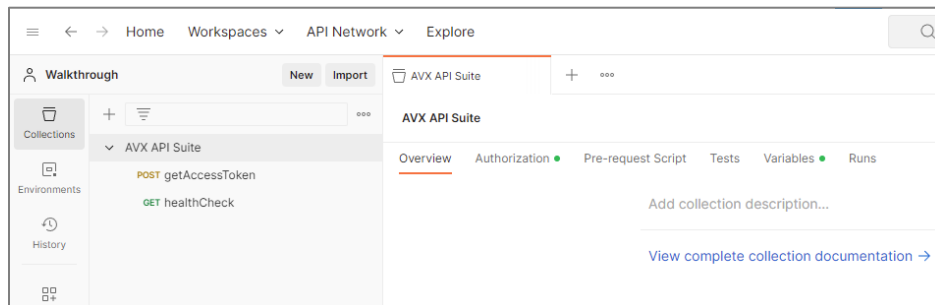
1. In Postman, click **Import**.



2. Click **files** and open the `AVX_API_Suite.postman_collection` file.
This imports the collection into your Postman client.



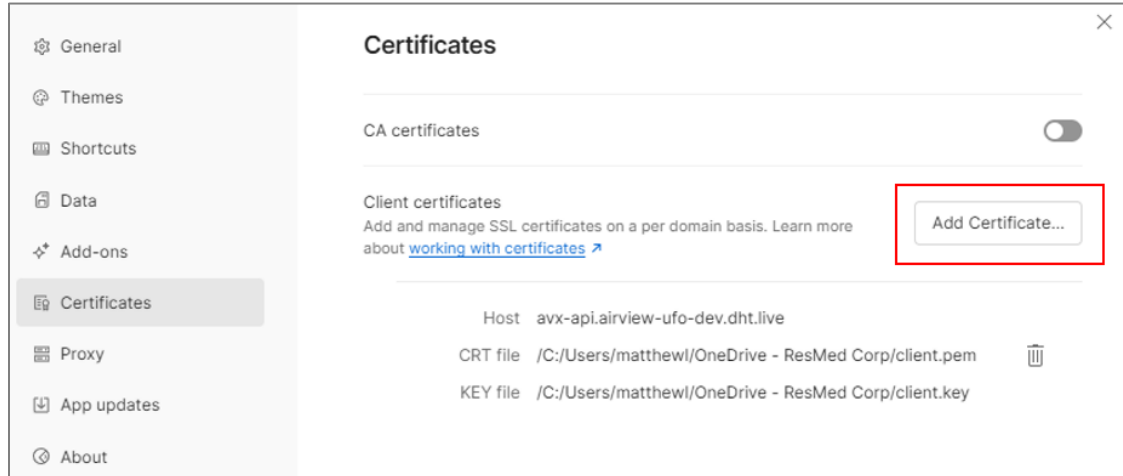
3. Click **AVX API Suite** to expand the full collection.
The full collection displays, and the configuration tab for the collection opens.



For additional instructions on Postman, click [here](#).

Import the certificate and certificate key

1. Click the **Settings** icon and select **Settings**.
2. Click **Certificates > Add Certificate**.



3. In the **Host** field, paste the AVX API URL (`api_base_url`), and import the `client.pem` file to the **CRT file** and the `client.key` file to the **KEY file**.

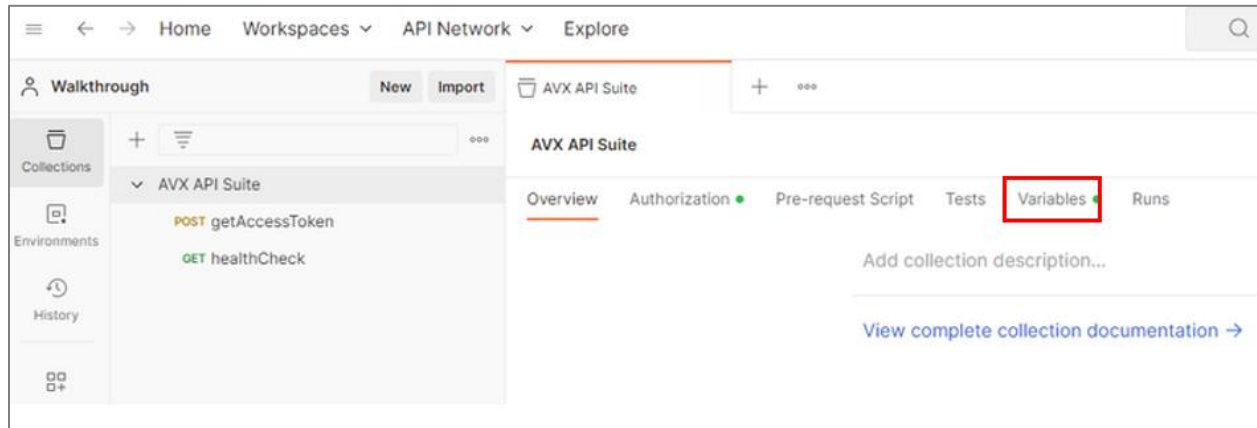
A screenshot of the 'Add certificate' dialog box in Postman. The dialog has a title bar and a close button (X) in the top right. It contains several input fields: 'Host (required)' with a text input containing 'https://' and a port input containing '443'; 'CRT file' with a text input containing 'client.pem' and a close button (X); 'KEY file' with a text input containing 'client.key' and a close button (X); 'PFX file' with a 'Select File' button; and 'Passphrase' with a text input and a toggle icon. At the bottom left are 'Add' and 'Cancel' buttons. At the bottom right is a link: 'Learn more about [working with certificates](#)'. The 'Add' button is highlighted in orange.

4. Click **Add**.

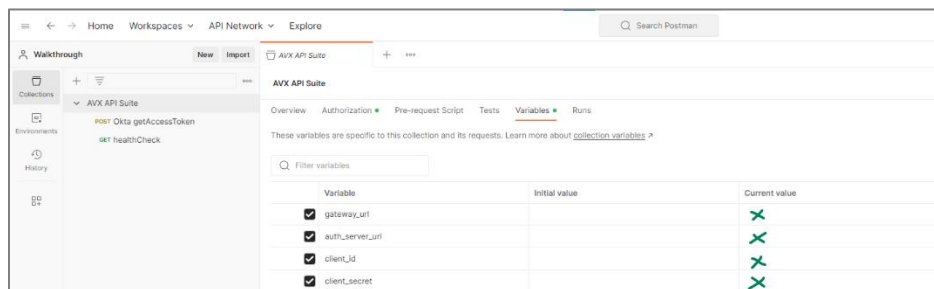
For more instructions on Postman, click [here](#).

Update the variables

1. On the **AVX API Suite** tab, click **Variables**.



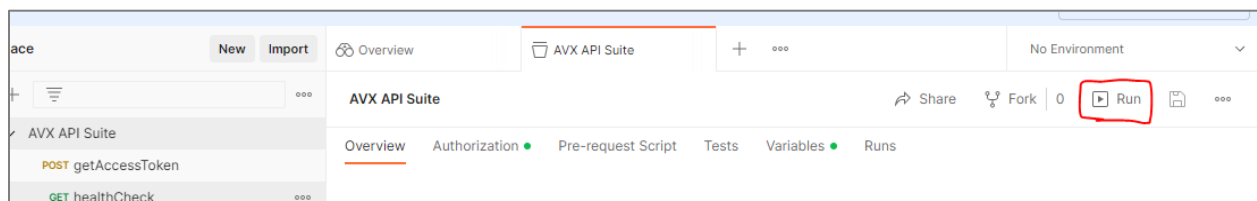
2. In the **Current** value column, in the **api_base_url** and **auth_server_url** fields, enter the AVX API URL and Authorization server URL, respectively. In the **client_id** and **client_secret** fields, paste your `client_id` and `client_secret`.



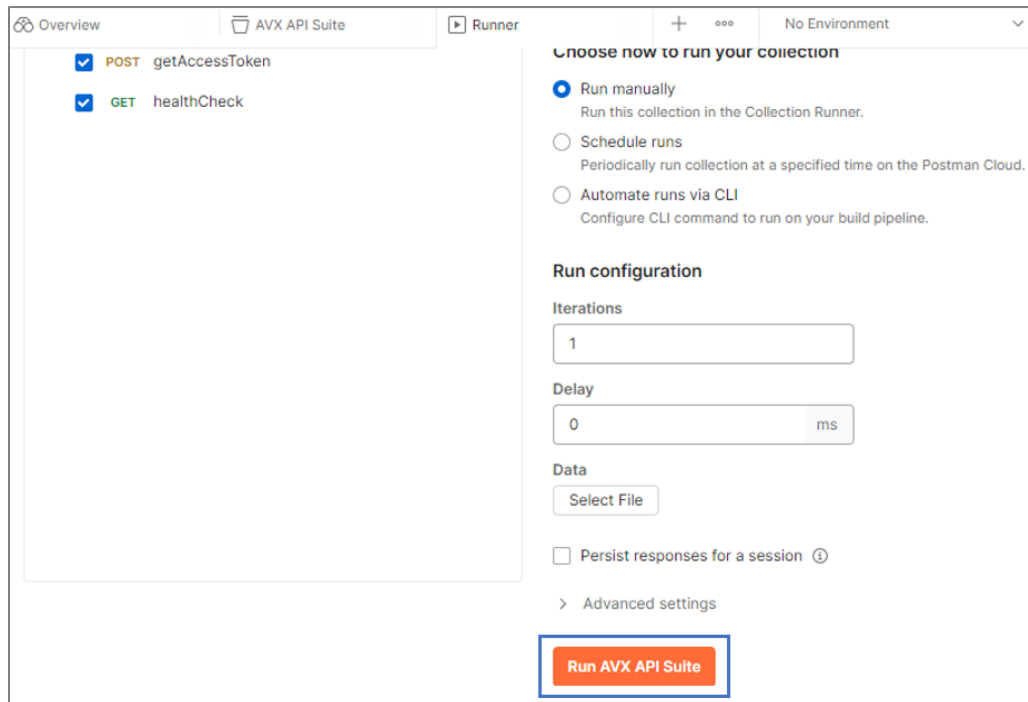
3. Click **Save**.

Run the API suite

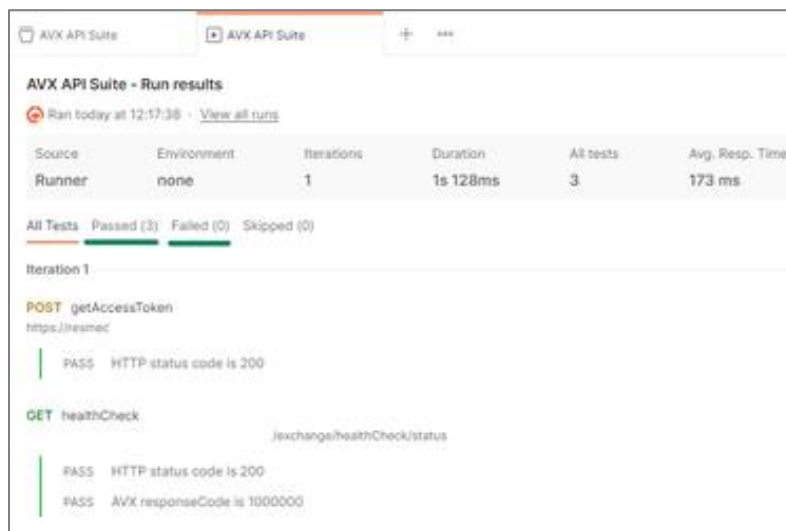
1. From the **AVX API Suite** tab, click **Run** on the right-hand side. This opens a new Run page.



2. On this new page, click **Run AVX API Suite** to execute the requests.

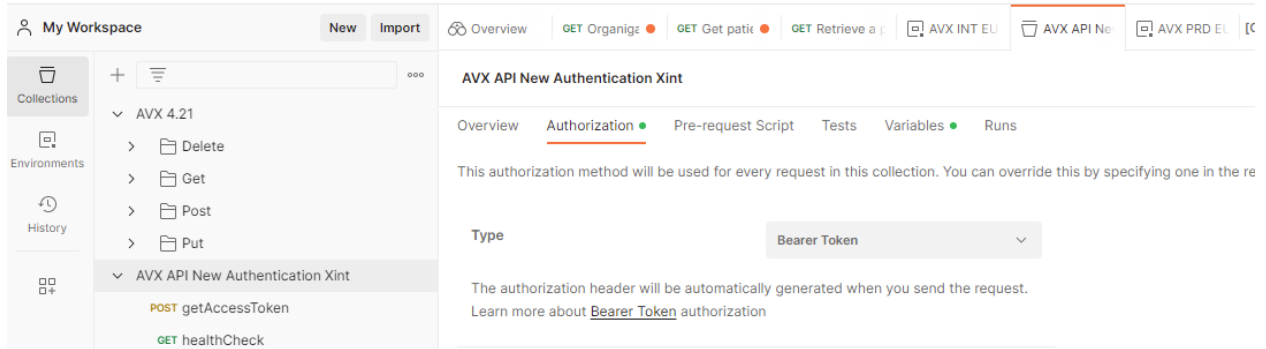


Postman makes requests on your behalf and validates that the onboarding successfully completed. A successful run should contain three passed tests and zero failed tests.



3. If any failures are present, consult with your customer representative.

Note: This should already be preconfigured in Postman, but in the Authorization configuration the type needs to be Bearer Token.



4. Certificate expiration and renewal process

Your certificate expires 365 days after it is created. An automated email from ResMed or other communication will come to you before the certificate expires to confirm that you want a new certificate and key. You are also welcome to reach out to ResMed as the certificate expiration date approaches to request a new certificate and key.

The following steps outline the process for renewing your certificate:

1. ResMed creates a new certificate and private key for you and uploads them to BlackBerry Workspaces. ResMed then sends the workspace link to your provided integration email address.
2. Update your integration to use the new certificate and private key before the expiration of your former certificate.

Note: ResMed can create or delete new certificates for you any time, but you can only have up to two active certificates at a given time.



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